Boundary value problems with conductors

$$\Phi(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\int \frac{\rho(\vec{r}\,')\,d^3\vec{r}\,'}{|\vec{r}-\vec{r}\,'|}$$

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\int \frac{(\vec{r}-\vec{r}\,')\rho(\vec{r}\,')}{|\vec{r}-\vec{r}\,'|^3}\,d^3\vec{r}\,'$$

$\rho(\vec{r})$ determines $\varphi(\vec{r})$ and $\vec{E}(\vec{r})$ fully.

But in <u>conductors</u> charges are free to move and their distribution is not known.

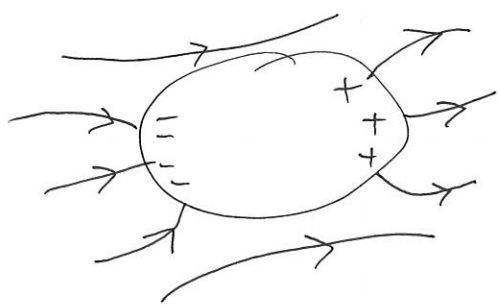
Since charges are at rest, $\vec{E}=0$ inside conductor $\Rightarrow \varphi = \text{const}$. Thus we can specify potential of the conductor (and maybe its total charge Q) but not $\rho(\vec{r})$.

- All charges are on the surface (otherwise $\nabla^2\Phi \neq 0$, $\vec{\nabla}\cdot\vec{E}\neq 0$)
- $\vec{E}$ just outside the surface is normal to the surface.

$$\oint \vec{E}\cdot\hat{n}\,dS = \frac{\sigma A}{\varepsilon_0} \Rightarrow \vec{E} = \frac{\sigma}{\varepsilon_0}\hat{n} = -\frac{\partial\varphi}{\partial n}\hat{n}$$

local surface charge density

$$\frac{\partial\varphi}{\partial n} = \hat{n}\cdot\vec{\nabla}\Phi.$$



Uniqueness of $\nabla^2\varphi(\vec{r}) = -\frac{\rho(\vec{r})}{\varepsilon_0}$ solution:

suppose two solutions φ_1, φ_2.

Consider $I = \int_V (\nabla\varphi_1 - \nabla\varphi_2)^2\,d^3\vec{r} = \int \vec{\nabla}(\varphi_2-\varphi_1)\cdot\vec{\nabla}(\varphi_2-\varphi_1)\,d^3\vec{r} =$

$$= \int \vec{\nabla}\cdot\left[(\varphi_2-\varphi_1)\,\vec{\nabla}(\varphi_2-\varphi_1)\right]d\vec{r} - \int (\varphi_2-\varphi_1)\underbrace{\nabla^2(\varphi_2-\varphi_1)}_{=0}\,d^3\vec{r}$$

$$\Rightarrow I = \sum_i \int_{S_i} (\varphi_2 - \varphi_1)\, \vec{\nabla}(\varphi_2 - \varphi_1)\cdot \hat{n}\, ds$$

sum over surfaces

On a given surface S_i $\varphi_2 - \varphi_1 = \text{const}$ (conductor is equipotential)

$$I = \sum_i (\varphi_2 - \varphi_1)_i \underbrace{\int_{S_i} \vec{\nabla}(\varphi_2 - \varphi_1)\cdot \hat{n}\, ds}_{-\int (\vec{E}_2 - \vec{E}_1)\cdot \hat{n}\, ds \,=\, -\frac{1}{\varepsilon_0}\int (\sigma_2 - \sigma_1)\, ds}$$

$$\Rightarrow I = -\frac{1}{\varepsilon_0} \sum_i (\varphi_2 - \varphi_1)_i (Q_2 - Q_1)_i =$$

$$= -\frac{1}{\varepsilon_0} \sum_i \big(\varphi_2(S_i) - \varphi_1(S_i)\big)\big(Q_2(S_i) - Q_1(S_i)\big)$$

potential on S_i — total charge on S_i

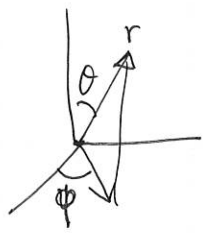
$\Rightarrow I = 0$, so that $\varphi_2 = \varphi_1$ (unique solution) for two types of boundary conditions:

1) total charge $Q(S_i) = Q_i$ on the i-th conductor is specified.

2) $\varphi(S_i) = \varphi_i$ potential on the i-th conductor is specified.

Separation of variables: in some symmetric situations the solution $\varphi(x,y,z) = F(q_1)G(q_2)H(q_3)$ separates into the product, $q_{1,2,3}$ are curvilinear coordinates.

Spherical coordinates:
$$x = r\sin\theta\cos\varphi$$
$$y = r\sin\theta\sin\varphi$$
$$z = r\cos\theta$$



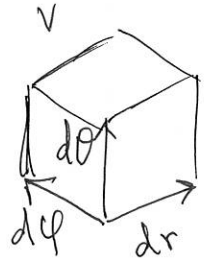
$$dx = \sin\theta\cos\varphi\, dr + r\cos\theta\cos\varphi\, d\theta - r\sin\theta\sin\varphi\, d\varphi$$
$$dy = \sin\theta\sin\varphi\, dr + r\cos\theta\sin\varphi\, d\theta + r\sin\theta\cos\varphi\, d\varphi$$
$$dz = \cos\theta\, dr - r\sin\theta\, d\theta$$

$$\Rightarrow dl^2 = dx^2 + dy^2 + dz^2 = h_r^2\, dr^2 + h_\theta^2\, d\theta^2 + h_\varphi^2\, d\varphi^2 \text{ (metric)}$$

$$h_r = 1, \quad h_\theta = r, \quad h_\varphi = r\sin\theta$$

Laplace $\nabla^2\phi = 0$ with $\phi(S_i)$ specified on the surface.

$$\Rightarrow \int_V d^3\vec{r}\, \nabla^2\phi = \int_V d^3\vec{r}\, \vec{\nabla}\phi\cdot\vec{\nabla}\phi = \oint_S \underbrace{\vec{\nabla}\phi\cdot\hat{n}}_{\frac{\partial\phi}{\partial n}}\, ds$$



Electric flux through the surface normal to r

$$dl_\varphi\, dl_\theta \frac{\partial\phi}{\partial l_r} = \frac{h_\varphi h_\theta\, d\varphi\, d\theta}{h_r} \frac{\partial\phi}{\partial r}$$

$\Rightarrow$ the net flux through this and opposite face is

$$\frac{h_\varphi h_\theta \, d\varphi d\theta}{h_r} \frac{\partial \phi}{\partial r}\bigg|_{r+dr} - \frac{h_\varphi h_\theta \, d\varphi d\theta}{h_r} \frac{\partial \phi}{\partial r}\bigg|_{r} =$$

$$= \frac{\partial}{\partial r}\left(\frac{h_\varphi h_\theta}{h_r} \frac{\partial \phi}{\partial r}\right) d\varphi d\theta dr$$

$\Rightarrow$ total flux through all ~~surfaces~~ faces

$$\left[\frac{\partial}{\partial r}\left(\frac{h_\varphi h_\theta}{h_r} \frac{\partial \phi}{\partial r}\right) + \frac{\partial}{\partial \theta}\left(\frac{h_r h_\varphi}{h_\theta} \frac{\partial \phi}{\partial \theta}\right) + \right.$$

$$\left. + \frac{\partial}{\partial \varphi}\left(\frac{h_r h_\theta}{h_\varphi} \frac{\partial \phi}{\partial \varphi}\right)\right] \underbrace{d\varphi d\theta dr}_{\frac{dV}{h_r h_\theta h_\varphi}} = 0$$

$$\frac{1}{h_r h_\theta h_\varphi}\left\{\frac{\partial}{\partial r}\left(r^2 \sin\theta \frac{\partial \phi}{\partial r}\right) + \frac{\partial}{\partial \theta}\left(\frac{r \sin\theta}{r} \frac{\partial \phi}{\partial \theta}\right) + \frac{\partial}{\partial \varphi}\left(\frac{r}{r\sin\theta} \frac{\partial \phi}{\partial \varphi}\right)\right\} = 0$$

~~Divide by $r\sin\theta$~~ $h_r h_\theta h_\varphi = r^2 \sin\theta$

$$\frac{1}{r^2 \sin\theta}\left\{\frac{\partial}{\partial r}\left(r^2 \cancel{\sin\theta} \frac{\partial \phi}{\partial r}\right) + \cancel{\frac{1}{\sin\theta}} \frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial \phi}{\partial \theta}\right) + \frac{1}{\sin\theta} \frac{\partial^2 \phi}{\partial \varphi^2}\right\} = 0$$

~~More carefully, element of the volume is~~

$$\frac{1}{r^2} \frac{\partial}{\partial r}\left(r^2 \frac{\partial \phi}{\partial r}\right) + \frac{1}{r^2 \sin\theta} \frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial \phi}{\partial \theta}\right)$$

$$+ \frac{1}{r^2 \sin^2\theta} \frac{\partial^2 \phi}{\partial \varphi^2} = 0$$

$$\Rightarrow \nabla^2 \phi = \frac{1}{r^2} \frac{\partial}{\partial r}\left(r^2 \frac{\partial \phi}{\partial r}\right) + \frac{1}{r^2 \sin\theta} \frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial \phi}{\partial \theta}\right)$$

$$+ \frac{1}{r^2 \sin^2\theta} \frac{\partial^2 \phi}{\partial \varphi^2}.$$

Can re-write as

$$\underbrace{\frac{\partial}{\partial r}\left(r^2 \frac{\partial \phi}{\partial r}\right)}_{2r\frac{\partial \phi}{\partial r} + r^2 \frac{\partial^2 \phi}{\partial r^2}} + \frac{1}{\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial \phi}{\partial \theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2 \phi}{\partial \varphi^2} = 0$$

$$r\frac{\partial^2}{\partial r^2}(r\phi) = r\frac{\partial}{\partial r}\left(\phi + r\frac{\partial \phi}{\partial r}\right) = r\left(\frac{\partial \phi}{\partial r} + \frac{\partial \phi}{\partial r} + r\frac{\partial^2 \phi}{\partial r^2}\right)$$

$\Rightarrow$ equivalent form is

$$r\frac{\partial^2}{\partial r^2}(r\phi) + \frac{1}{\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial \phi}{\partial \theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2 \phi}{\partial \varphi^2} = 0$$

Then can assume separable solution of the form

$$\phi(r,\theta,\varphi) = \frac{R(r)}{r}\, U(\cos\theta)\, F(\varphi)$$

$$r\frac{\partial^2}{\partial r^2}(R)\, UF + \frac{1}{\sin\theta}\frac{\partial}{\partial \theta}(\sin\theta\, U)\, RF +$$

$$+ \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial \varphi^2}(F)\, RU = 0$$

Divide by $\phi = \frac{R}{r} UF$

$$\frac{r}{R}\frac{d^2R}{dr^2} + \frac{1}{U\sin\theta}\frac{d}{\partial\theta}(\sin\theta\, U) + \frac{1}{\sin^2\theta\, F}\,\partial$$

$$r \frac{d^2}{dr^2}(R)\, UF + \frac{1}{\sin\theta} \frac{d}{d\theta}\left(\sin\theta \frac{dU}{d\theta}\right) \frac{RF}{r} +$$

$$+ \frac{1}{\sin^2\theta} \frac{d^2 F}{d\varphi^2} \frac{RU}{r} = 0$$

Divide by $\phi = \frac{RUF}{r}$

$$\frac{r^2}{R} \frac{d^2 R}{dr^2} + \frac{1}{U \sin\theta} \frac{d}{d\theta}\left(\sin\theta \frac{dU}{d\theta}\right) + \frac{1}{F \sin^2\theta} \frac{d^2 F}{d\varphi^2} = 0$$

let $u = \cos\theta$, $\sin^2\theta = 1 - u^2$

$$\frac{d}{d\theta} = \frac{d}{du} \frac{du}{d\theta} = -\sin\theta \frac{d}{du} \Rightarrow \sin\theta \frac{d}{d\theta} = -\sin^2\theta \frac{d}{du}$$

$$\frac{1}{\sin\theta} \frac{d}{d\theta}\left(\sin\theta \frac{dU}{d\theta}\right) = \frac{1}{\sin\theta}\left(-\sin\theta \frac{d}{du}\right)\left(-\sin^2\theta \frac{dU}{du}\right)$$

$$= \frac{d}{du}\left((1-u^2) \frac{dU}{du}\right)$$

$$\frac{r^2}{R} \frac{d^2 R}{dr^2} + \frac{1}{U}\left(\frac{d}{du}(1-u^2) \frac{dU}{du}\right) + \frac{1}{(1-u^2)F} \frac{d^2 F}{d\varphi^2} = 0$$

$$\frac{1}{F} \frac{d^2 F}{d\varphi^2} = -m^2$$ because other terms in the equation are φ-independent

$\Rightarrow F \sim e^{im\varphi}$

$$\Rightarrow \underbrace{\frac{r^2}{R} \frac{d^2 R}{dr^2}}_{\text{only depends on } r} + \underbrace{\frac{1}{U} \frac{d}{du}\left[(1-u^2) \frac{dU}{du}\right] - \frac{m^2}{1-u^2}}_{\text{only depends on } u} = 0$$

$$\Rightarrow \quad \frac{r^2}{R}\frac{d^2R}{dr^2} = \text{const} \equiv l(l+1)$$

$$\Rightarrow \begin{cases} \dfrac{d}{du}\left[(1-u^2)\dfrac{dU}{du}\right] + \left[l(l+1) - \dfrac{m^2}{1-u^2}\right]U = 0. \\[2ex] \dfrac{d^2R}{dr^2} = l(l+1)\dfrac{R}{r^2} \\[2ex] \dfrac{d^2F}{d\varphi^2} = -m^2 F \quad , \; F \sim e^{im\varphi} \text{ with integer } m. \end{cases}$$

$$\frac{d^2R}{dr^2} = l(l+1)\frac{R}{r^2} \quad : \text{ try } R = r^\nu$$

$$\frac{d^2 r^\nu}{dr^2} = \nu\frac{dr^{\nu-1}}{dr} = \nu(\nu-1)\, r^{\nu-2}$$

$$\frac{R}{r^2} = r^{\nu-2}$$

$$\Rightarrow \quad \nu(\nu-1)\, r^{\nu-2} = l(l+1)\, r^{\nu-2}$$

$$\nu^2 - \nu = l^2 + l \Rightarrow$$

$$\Rightarrow \quad \nu^2 - \nu - l(l+1) = 0$$

$$\nu = \frac{1 \pm \sqrt{1+4l(l+1)}}{2} = \frac{1 \pm \sqrt{4l^2+4l+1}}{2} =$$

$$= \frac{1 \pm \sqrt{(2l+1)^2}}{2} = \frac{1 \pm (2l+1)}{2} = \begin{cases} l+1 \\ -l \end{cases}$$

Thus $R(r) = A r^{l+1} + B r^{-l}$

or $\frac{R}{r} = A r^{l} + \frac{B}{r^{l+1}}$.

The U-equation: for $\underline{m=0}$ we get

$$\frac{d}{du}\left((1-u^2)\frac{dU}{du}\right) + l(l+1)\,U = 0 \qquad \text{Legendre equation}$$

solved by Legendre polynomials $P_l(u)$:

$$P_0(u) = 1, \quad P_1(u) = u, \quad P_2(u) = \frac{1}{2}(3u^2-1) \ldots$$

$$P_l(u) = \frac{1}{2^l l!}\frac{d^l}{du^l}(u^2-1)^l.$$

For $m \neq 0$ $\quad \frac{d}{du}\left((1-u^2)\frac{dU}{du}\right) + \left(l(l+1) - \frac{m^2}{1-u^2}\right)U = 0$

associated Legendre polynomials

$$P_l^m(u) = \frac{(-1)^m}{2^l\, l!}(1-u^2)^{m/2}\frac{d^{l+m}}{du^{l+m}}(u^2-1)^l \quad (m \leq l)$$

The full angular dependence is described by the <u>spherical harmonics</u>

$$Y_{lm}(\theta,\varphi) = \sqrt{\frac{2l+1}{4\pi}\frac{(l-m)!}{(l+m)!}}\; P_l^m(\cos\theta)\, e^{im\varphi}, \quad -l \leq m \leq l$$

$$Y_{lm}^*(\theta,\varphi) = (-1)^m\, Y_{l,-m}(\theta,\varphi).$$

Orthogonality: $\int_0^{2\pi} d\varphi \int_0^{\pi} d\theta \sin\theta\; Y_{lm}^*(\theta,\varphi)\, Y_{l',m'}(\theta,\varphi) = \delta_{ll'}\,\delta_{mm'}$

$\Rightarrow$ any function on the sphere is represented by

$$f(\theta,\varphi) = \sum_{l=0}^{\infty} \sum_{m=-l}^{l} a_{lm} Y_{lm}(\theta,\varphi)$$

$$a_{lm} = \int_0^{2\pi} d\varphi \int_0^{\pi} d\theta \sin\theta \; Y_{lm}^*(\theta,\varphi) \; f(\theta,\varphi).$$

P 3.15 Conducting sphere of radius R_0 with potential $\phi = 0$ is placed in uniform electric field

$\phi = -E_0 z$.

Find ϕ outside of it.

$\nabla^2 \phi = 0$ and problem is azimuthally-symmetric (rotation about $\hat{z}$ = field axis).

$$\Rightarrow \phi(r,\theta,\varphi) = \sum_{\ell,m} \left(A_{\ell m} r^\ell + \frac{B_{\ell m}}{r^{\ell+1}} \right) Y_{\ell m}(\theta,\varphi) \xrightarrow[m=0]{}$$

$$\rightarrow \sum_\ell \left(A_\ell r^\ell + \frac{B_\ell}{r^{\ell+1}} \right) P_\ell(\cos\theta) \quad \text{for } r \geq R_0.$$

Note that potential of the uniform field $-E_0 z = -E r \cos\theta$

It describes $A_\ell r^\ell P_\ell(\cos\theta) = A_1 r \cos\theta$ with $A_1 = -E_0$.

Boundary condition: $\phi(R_0, \theta) = 0 \Rightarrow A_1 R_0 + \frac{B_1}{R_0^2} = 0$

$$\Rightarrow B_1 = -A_1 R_0^3 = E_0 R_0^3.$$

$$\Rightarrow \phi(r,\theta) = -E_0 r \cos\theta + \frac{E_0 R_0^3}{r^2} \cos\theta$$

(all other coefficients, $A_{\ell \neq 1} = 0 = B_{\ell \neq 1}$).

Note that $\frac{E_0 R_0^3}{r^2} \cos\theta$ is potential of electric dipole

$$\frac{E_0 R_0^3}{r^2} \cos\theta = \frac{1}{4\pi\varepsilon_0} \frac{\vec{R} \cdot \vec{d}}{r^3} \Rightarrow \vec{d} = 4\pi\varepsilon_0 R_0^3 E_0 \hat{z}.$$

$$\vec{E} = -\vec{\nabla}\phi = -\left(\frac{\partial\phi}{\partial r}\hat{r} + \frac{1}{r}\frac{\partial\phi}{\partial\theta}\hat{\theta} + \frac{1}{r\sin\theta}\frac{\partial\phi}{\partial\varphi}\hat{\varphi}\right)$$

$$\Rightarrow E_r = +E_0\cos\theta + \frac{2E_0R_0^3}{r^3}\cos\theta = E_0\cos\theta\left(1+\frac{2R_0^3}{r^3}\right)$$

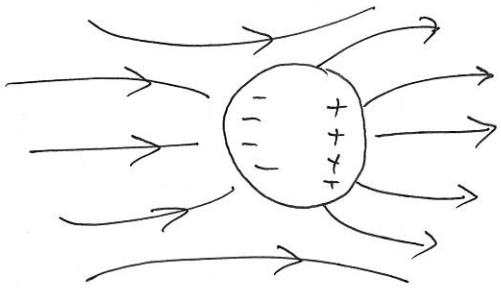
$$E_\theta = -\frac{\partial}{\partial\theta}\left(-E\cos\theta + \frac{E_0R_0^3}{r^3}\cos\theta\right) = -E_0\sin\theta\left(1-\frac{R_0^3}{r^3}\right)$$

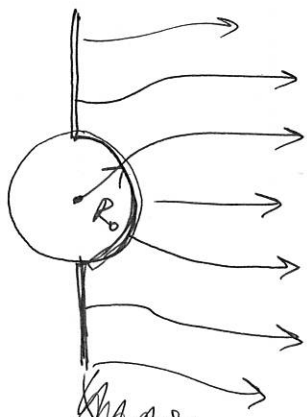
$$E_\varphi = 0.$$

$\vec{E}$ reaches max at the poles $(\theta = 0, \pi)$ on the surface of the sphere where $\vec{E} = E_r\hat{r} = 3\vec{E}_0$
(el. field enhancement at the extremes of the surface is a very general phenomenon).

Surface charge density $\sigma = \varepsilon_0\frac{\partial\phi}{\partial n} = \varepsilon_0 E_r\Big|_{r=R_0} =$

$$= 3\varepsilon_0 E_0\cos\theta$$



Another interesting observation: $\phi(r,\theta) = E_0 r\cos\theta\left(\frac{R_0^3}{r^3}-1\right)$ is 0 at $\theta = \pi/2$.

Therefore this ϕ is also the potential of the infinite charged conducting surface with spherical semi-sphere protrusion of Radius R_0.